



MIT Art, Design and Technology University

MIT School of Computing, Pune

Department of Computer Science and Engineering

Assignments

Course – Agentic AI

Class - L.Y. (SEM-I), AIA

Name of the Course Coordinator

Dr. Swapnili Karmore

Team Members

A.Y. 2026 - 2027

ASSIGNMENT	
Assignment No: 02	Unit: Unit II -- Reasoning, Planning and Decision-Making
Title: ReAct Agent: Chain-of-Thought vs Tree-of-Thought Reasoning for Autonomous Task Planning	
Mapped Course Outcome:	CO2 -- Design and implement reasoning and planning mechanisms for AI agents. (Apply / Analyze)
Objective:	To enable students to implement and compare Chain-of-Thought, Tree-of-Thought, and ReAct (Reason + Act) reasoning strategies for goal decomposition and autonomous task execution, including handling of uncertainty and dynamic replanning.
Problem Statement:	Using an LLM API, build an agent that decomposes a multi-step real-world task (e.g., planning a 3-day trip on a fixed budget, or debugging a failing script) using three reasoning strategies -- Chain-of-Thought, Tree-of-Thought, and the ReAct framework -- and compare their planning quality, efficiency, and robustness when a constraint changes mid-task.
Implementation Tasks:	1. Select a multi-step planning task and clearly define its goal, constraints, and available tools/actions.
	2. Implement Chain-of-Thought prompting for step-by-step reasoning and record the generated plan.
	3. Implement Tree-of-Thought reasoning that generates and evaluates multiple candidate plan branches before selecting the best one.
	4. Implement the ReAct framework, interleaving Reason and Act steps with at least one external tool call (calculator, search, or mock API).
	5. Introduce a mid-task disruption (a changed constraint or sub-goal) and test each strategy's dynamic replanning capability.
	6. Log and compare the number of reasoning steps, task success rate, time/tokens used, and quality of the final plan for each strategy.
	7. Prepare a 1-page report comparing CoT vs ToT vs ReAct with guidance on when each is preferable.
Tools & Technologies:	Python, an LLM API (OpenAI / Gemini / Groq or an open-source model), LangChain (optional), Jupyter/Google Colab.
Expected Deliverables:	- Notebook implementing Chain-of-Thought, Tree-of-Thought, and ReAct agents on the same task. - Log/table of the dynamic-replanning test results for all three strategies. 1-page comparison report with recommendations on strategy selection. - Live demo showing the agent replanning in real time after a disruption.
Evaluation Criteria	
Criterion	Marks
Chain-of-Thought reasoning implementation and plan quality	2
Tree-of-Thought reasoning implementation and branch evaluation	2
ReAct framework implementation with tool-call integration	2
Dynamic replanning test and robustness analysis	2

Comparison report and live demo/viva responses	2
Total	10
Reference / Resources:	• Yao et al., ReAct: Synergizing Reasoning and Acting in Language Models • Wei et al., Chain-of-Thought Prompting Elicits Reasoning in Large Language Models • Unit II syllabus notes -- Reasoning, Planning and Decision-Making
Faculty Signature:	HOD Signature:

Prepared By

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Note:

In last column L1 is Blooms Taxonomy Level and CO2 Course Outcome. Write Blooms Taxonomy Level and Course Outcome in front of each question in short-hand notation